IBM Db2 Overview, Architecture & Use Cases DBMS Module 6

> **Definition: IBM Db2** is an enterprise-grade Relational Database Management System (RDBMS) developed by IBM, engineered for high-performance transactional processing (OLTP), analytical workloads (OLAP), and hybrid data management (HTAP) across hybrid cloud environments.

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1. Detailed Technical Explanation

1. Key Features of IBM Db2:

- **BLU Acceleration:** In-memory columnar processing for multi-fold speedup in analytical queries.

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2. IBM Db2 Product Editions

| Db2 Edition | Target Environment | Hardware / Resource Limits | Key Features |

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3. Industry Use Cases

1. **Banking & Financial Services:** Core banking transactions, real-time credit card fraud detection using pureScale high availability.

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4. Comparison: IBM Db2 vs Other RDBMS

| Feature | IBM Db2 | Oracle Database | MySQL / PostgreSQL |

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5. Core Concepts & Memory Keywords

- **pureScale:** IBM active-active database clustering for zero downtime.

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6. Must-Write Points for Exams

- IBM Db2 supports Hybrid Transactional and Analytical Processing (HTAP) in a single engine.

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7. Quick Recall Flow

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